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Biostatistics deserts and scholarly productivity among US medical schools

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The teaching and practice of biostatistics are essential to medical education, but access to biostatistical expertise is limited at many medical schools. Medical schools in rural and historically underserved areas may be doubly disadvantaged in accessing biostatistics expertise by their lack of financial resources and a lack of local programmes training future generations of biostatisticians. Using public data on US medical schools and biostatistics PhD programmes, we identified medical schools operating in 'biostatistics deserts' (institutions without an affiliated or colocated PhD programme in biostatistics) and correlated each medical school's location in a biostatistics desert with scholarly productivity, operationalized as the annual number of scholarly publications. Among 126 MD-granting medical schools in our analysis, 46% were located in a biostatistics desert and had a median of 590 publications/year, compared to 993 for medical schools with a colocated PhD programme and 1,369 with an affiliated programme. On multivariable analysis, the presence of a Biostatistics, Epidemiology, and Research Design (BERD) programme, but not affiliation or colocation with a biostatistics PhD program, was associated with higher scholarly productivity. Structured biostatistics services, such as BERD programmes, may represent the best opportunity for medical schools to leverage the local biostatistics workforce to support scholarly publication.

1 | INTRODUCTION

With the increasing methodological complexity of biomedical research, the teaching and practice of biostatistics have become essential to the educational and research missions of medical schools and academic medical centres (AMCs) (Welty et al., 2013). However, medical trainees frequently cite insufficient access to biostatistics expertise: in 2022, almost 3 in 10 graduating physicians described their preclinical training in biostatistics and epidemiology as fair or poor (Association of American Medical Colleges, 2022a), and surveys of resident physicians revealed that approximately half consider a lack of biostatistics knowledge or the absence of biostatistical support as barriers to participating in scholarly activity (Abramson et al., 2018; Mansi et al., 2019). The shortage of biostatisticians is a fundamental cause of limited biostatistics support for medical trainees, and this shortage begins with the biostatistician training pipeline. For example, in 2021, only 222 doctoral degrees in biostatistics were awarded in the United States (American Statistical Association, n.d.-b), as compared to 28,337 MD and DO degrees (Kaiser Family Foundation, n.d.). While some medical schools have sought to address this shortage by leveraging the expertise of master's-level biostatisticians and PhDs in related fields (Desai et al., 2022; McKinney et al., 2019), the rarity of biostatistics PhDs relative to physicians is emblematic of a nationwide imbalance between biostatistician supply and demand.

The nationwide shortage of biostatisticians is exacerbated by inequality among medical schools in access to biostatistics resources. In 2010–2013, a quarter of US medical schools had four or fewer biostatistics faculty, and a quarter of 'small' schools (those with <700 faculty) had just a single biostatistician faculty member (Zhang & Chen, 2015). While some medical schools and AMCs benefit from in-house biostatistics cores funded by the National Institutes of Health (NIH) (Hanlon et al., 2022), such cores are absent at many medical schools due to the highly unequal

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distribution of NIH funding across institutions. Medical schools in rural and historically underserved areas may be doubly disadvantaged in accessing biostatistics expertise because of their lack of financial resources (e.g., NIH and other funding for biostatistics cores) and a lack of local programmes training future generations of biostatisticians. We refer to the latter disadvantage as the phenomenon of 'biostatistics deserts', mirroring the concept of 'food deserts' that describes areas lacking in retailers of nutritious foods (Testa et al., 2021).

Medical schools operating in a biostatistics desert may encounter several barriers to teaching and research involving biostatistical methods. First, these schools may encounter difficulty recruiting biostatistics faculty and staff due to the absence of a local talent pool. Second, any biostatisticians who are recruited are at risk of isolation, which could lead to lower productivity, career dissatisfaction, and departure from the institution (Mehta et al., 2022). Third, schools in biostatistics deserts would be unable to draw upon support from biostatistics trainees (e.g., biostatistics student interns and student-led consulting services) (Kaseda et al., 2022; Pomann et al., 2022). Altogether, these barriers may exacerbate disparities among US medical schools in scholarly productivity, as described in prior research (Gray et al., 2023). In this study, we aimed to define how many US medical schools are located in a biostatistics desert, defined by the absence of a colocated or affiliated biostatistics training program. Our secondary aim was to describe the correlation between biostatistics deserts and medical schools' research productivity, measured through the number of scholarly publications.

2 | METHODS

This study did not include human subjects research and did not require Institutional Review Board approval. We included US-based MD-granting medical schools, identified by querying the Association of American Medical Colleges, AAMC (n.d.) website. We excluded regional campuses, which may have had access to biostatistics support through their affiliation with the flagship campus of the same school. Following prior research (Gray et al., 2023; Tumin, Khanchandani, et al., 2022), we also excluded medical schools not uniquely identifiable by affiliation search in the Scopus database, as well as schools missing data on any other study variables (including schools not represented in the Blue Ridge Institute for Medical Research [BRIMR] summary of NIH funding).

For each medical school, the primary outcome was the availability of a local biostatistics training programme. This was defined based on a directory of graduate programmes maintained by the American Statistical Association, ASA (n.d.-a). Our primary definition of this outcome was based on the availability of a PhD programme in biostatistics. In secondary analyses, we expanded this definition to include (a) PhD programmes in either statistics or biostatistics and (b) master's or PhD programmes in either statistics or biostatistics. For each definition, we classified medical schools as having (a) an affiliated training programme (housed in the same institution as the medical school); (b) a colocated training programme (housed in a different institution, but located in the same county as the medical school's flagship campus); or (c) neither. Medical schools with neither an affiliated nor a colocated biostatistics training programme were considered to be located in a biostatistics desert. Our secondary outcome was each medical school's number of scholarly publications in 2022, determined by a search of the Scopus database.

Covariates in our analysis included each school's public or private designation, geographic region (based on US Census categories), and number of full-time faculty as of December 2022 (Association of American Medical Colleges, 2022b). We also controlled for total funding received in fiscal year 2022 from the NIH (recoded to $50 million dollar increments), as compiled by Blue Ridge Institute for Medical Research (BRIMR, 2023). Lastly, we controlled for the presence of Biostatistics, Epidemiology and Research Design (BERD) programmes associated with Clinical and Translational Science Awards (CTSA) programme funding through the NIH's National Center for Advancing Translational Sciences.

Continuous data were summarized using medians with interquartile ranges, while categorical data were summarized using counts with percentages. We grouped medical schools based on access to biostatistics training programmes (those with affiliated biostatistics training programs, those with colocated programmes and those with no local programs) to describe prevalence and geographic clustering of medical schools in biostatistics deserts. We compared study covariates and access to biostatistics training programmes using chi-square tests, Fisher's exact tests or Wilcoxon/Kruskal–Wallis tests, as appropriate. We compared the number of publications across the categories of medical school access to biostatistics training programmes, as well as individual study covariates, using negative binomial regression (due to overdispersion and skewness in distribution), with associations quantified using incident rate ratios (IRRs) with 95% confidence intervals (95% CIs). Lastly, we fit a multivariable negative binomial regression model for the number of publications including access to biostatistics training programmes and study covariates. Collinearity was assessed prior to creating multivariable models. Data analysis was performed in JMP Pro 17 (SAS Institute Inc., Cary, NC), and $P < .05$ was considered statistically significant.

3 | RESULTS

We identified 155 medical schools on the AAMC list, of which we excluded 9 schools that could not be uniquely identified in the Scopus database and 20 schools that were not included in the BRIMR rankings, resulting in a sample of $n = 126$. Table 1 reports descriptive statistics for the

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TABLE 1 Sample description for included medical schools ($n = 126$)

Measure	Median [IQR] or counts (%)
Total Faculty	1,057 [1,138]
Type	
Public	79 (62.7%)
Private	47 (37.3%)
US region	
Southern	46 (36.5%)
Northeast	33 (26.2%)
Central	28 (22.2%)
Western	19 (15.1%)
NIH Total funding	$68,200,000 [$189,700,000]
BERD programme	60 (47.6%)
Institutional publications	808 [1829]
Biostatistics PhD programme	
Affiliated	53 (42.1%)
Colocated	15 (11.9%)
None	58 (46.0%)
Biostatistics or statistics PhD programme	
Affiliated	74 (58.7%)
Colocated	14 (11.1%)
None	38 (30.2%)
Biostatistics or statistics PhD or master's programme	
Affiliated	92 (73.0%)
Colocated	15 (11.9%)
None	19 (15.1%)

Note: Colocated→ located in same county as an institution with a training programme.
Abbreviations: BERD, Biostatistics, Epidemiology, and Research Design; IQR, interquartile range; NIH, National Institutes of Health.

sample. Nearly two thirds of schools were public, with the fewest schools located in the Western United States. Nearly half of schools (48%) had a BERD programme. Median NIH funding for schools was $68 million, and median total number of school publications in 2022 was 808.

Nearly half (46%) of medical schools had neither an affiliated nor a colocated biostatistics PhD programme. Figure 1 depicts maps of biostatistics PhD programmes affiliated or colocated with medical schools. New York had the highest concentration of medical schools with affiliated biostatistics PhD programmes, and half of medical schools in the Northeastern United States (52%) had an affiliated biostatistics PhD programme (Table 2). Conversely, most medical schools in the Southern (54%) and Western (63%) United States had neither an affiliated nor a colocated biostatistics PhD programme. Medical schools with an affiliated biostatistics PhD programme were larger and had more NIH funding, having nearly double the amount of faculty and over seven times the funding as medical schools without affiliated biostatistics PhD programmes (Table 2). Over half (57%) of public medical schools did not have either an affiliated or a colocated biostatistics PhD program, compared to only 27% of private schools. Finally, a majority of BERD programmes (65%) were at medical schools with an affiliated biostatistics PhD programme. Table S1 summarizes these variables for schools with biostatistics or statistics PhD programmes and for schools with biostatistics or statistics PhD or master's programmes.

In unadjusted analysis (Table 3), the presence of an affiliated biostatistics PhD programme, compared to schools with no programme, was associated with higher school-wide publications (IRR = 2.6, 95% CI: 1.7–3.8; Figure 2). A similar association was found for schools with colocated programmes (IRR = 2.2, 95% CI: 1.2–4.0; Figure 2). Presence of a BERD programme (IRR = 2.6, 95% CI: 1.7–3.8), private schools (IRR = 2.6, 95% CI: 1.7–3.8), location in Northeastern United States (IRR = 2.6, 95% CI: 1.7–3.8) and higher NIH funding (IRR per $50 million increase = 1.3, 95% CI: 1.02–1.4) were all also associated with higher school-wide publications in unadjusted analyses (Figure 3). However, in adjusted analyses, the association between biostatistics PhD programmes and number of publications was completely attenuated. Meanwhile, the presence of BERD programs, US Census region and NIH funding remained significantly associated with school-wide publications in the adjusted model. We ran an additional analysis that did not include NIH funding as a covariate (Table S2), with presence of BERD remaining the strongest independent

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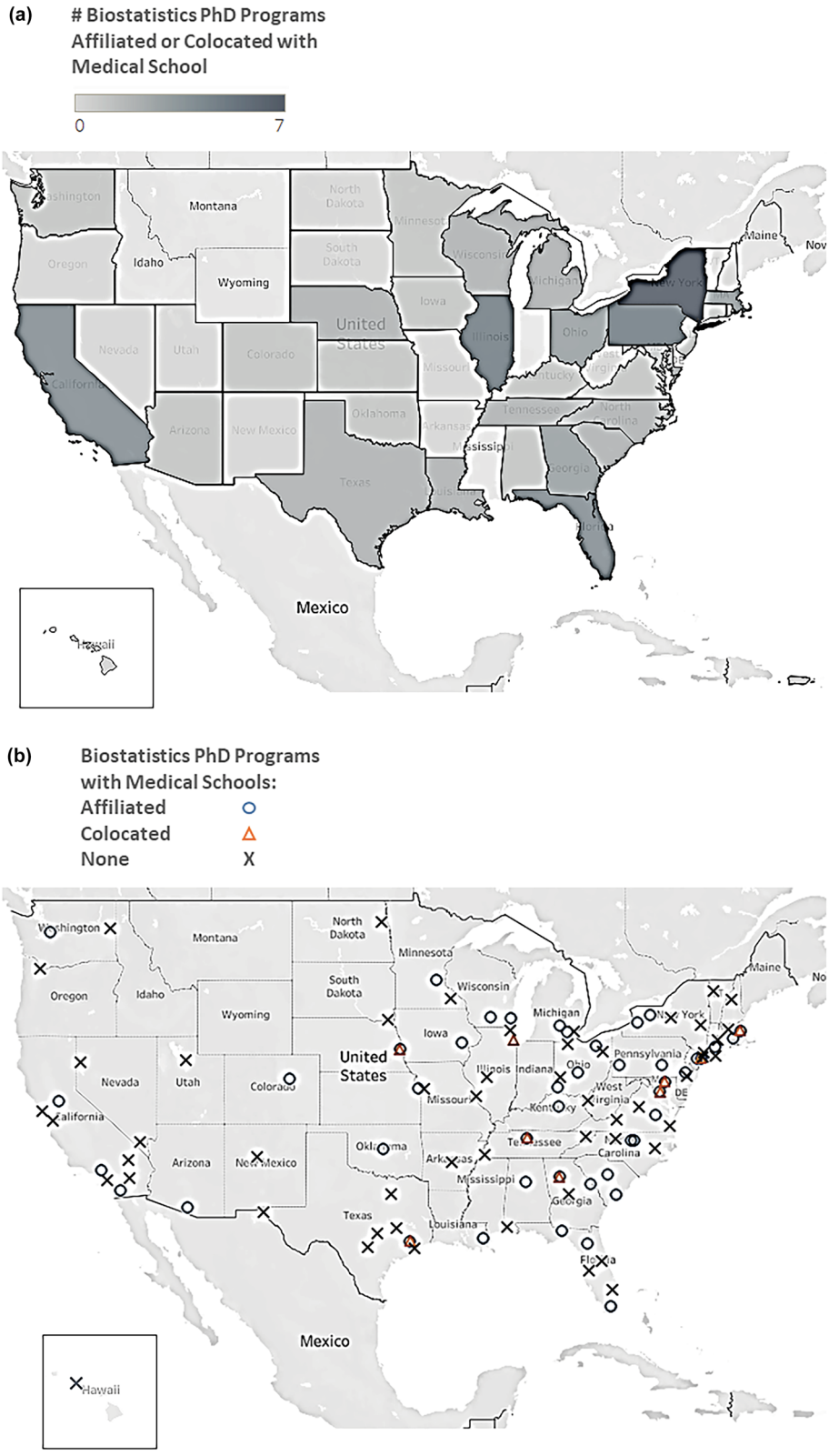


FIGURE 1 Density (a) and location (b) maps of medical schools with affiliated and colocated biostatistics PhD programmes.

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TABLE 2 Association between institutional characteristics and presence of an affiliated or colocated biostatistics PhD programme.

Measure	Affiliated programme ($n = 53$)	Colocated programme ($n = 15$)	No programme ($n = 58$)	*P* value
Total Faculty, median [IQR]	1,494 [1,313]	1,030 [1,619]	752 [845]	<.001
Type, counts (%)				
Public	31 (39.2%)	3 (3.8%)	45 (57.0%)	<.001
Private	22 (46.8%)	12 (25.5%)	13 (27.7%)	
US region				.070
Southern	18 (39.1%)	3 (6.5%)	25 (54.4%)	
Northeast	17 (51.5%)	6 (18.2%)	10 (30.3%)	
Central	11 (39.3%)	6 (21.4%)	11 (39.3%)	
Western	7 (36.8%)	0 (0%)	12 (63.2%)	
NIH Total funding, median [IQR]	$138,800,000 [298,500,000]	$56,300,000 [263,800,000]	$18,800,000[81,200,000]	<.001
BERD programme, counts (%)	39 (65.0%)	5 (3.3%)	16 (27.6%)	<.001

Note: *P* values were from chi-square or Fisher's exact tests for categorical measures and Wilcoxon/Kruskal–Wallis tests for continuous measures.
Abbreviations: BERD, Biostatistics, Epidemiology, and Research Design; NIH, National Institutes of Health.

TABLE 3 Incident rate ratios, with 95% confidence intervals for associations between presence of an affiliated or colocated biostatistics PhD programme, as well as institutional characteristics, with institutional publications.

Biostatistics PhD programme	Unadjusted IRR 95% CI	Adjusted IRR
Affiliated vs. no programme	**2.6** **1.7–3.8**	1.2 0.9–1.7
Colocated vs. no programme	**2.2** **1.2–4.0**	1.3 0.8–2.0
BERD programme (yes vs. no)	**4.7** **3.4–6.4**	**2.0** **1.4–2.8**
Type of programme (private vs. public)	**2.8** **1.8–4.1**	1.2 0.9–1.7
Region (northeast US vs. other)	**2.2** **1.4–3.3**	**1.7** **1.2–2.4**
NIH funding (per $50 million increase in funding)	**1.3** **1.2–1.4**	**1.2** **1.1–1.2**

Note: Adjusted IRRs from multivariable negative binomial regression including all variables in table. Statsitically significant associations in **bold**.
Abbreviations: BERD, Biostatistics, Epidemiology, and Research Design; CI, confidence interval; IRR, incident rate ratios; NIH, National Institutes of Health.

predictor of publications. Similar results were found when analysing access to biostatistics or statistics PhD programmes and when analysing access to biostatistics or statistics PhD or master's programmes (Tables S3 and S4).

4 | DISCUSSION

This study demonstrates the existence of biostatistics deserts affecting nearly half of US-based MD-granting medical schools. Biostatistics deserts are associated with geographical region, medical school size, private or public affiliation and NIH funding. Medical school affiliation or colocation with a biostatistics PhD programme was associated with much greater school-wide publication productivity in bivariate analysis, although the association between affiliated biostatistics PhD programmes and publications was attenuated after accounting for the presence of a BERD programme. This suggests that the mere presence of a biostatistics training programme is not sufficient to support biomedical research and that there needs to be group of biostatistical and research experts with a focus and understanding of biomedical research to best support clinical faculty research endeavours.

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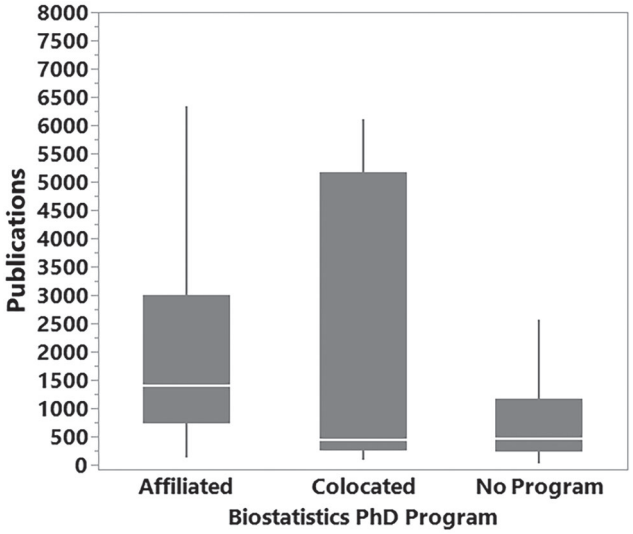


FIGURE 2 Box plot of publications for medical schools based on affiliation with biostatistics PhD programme.

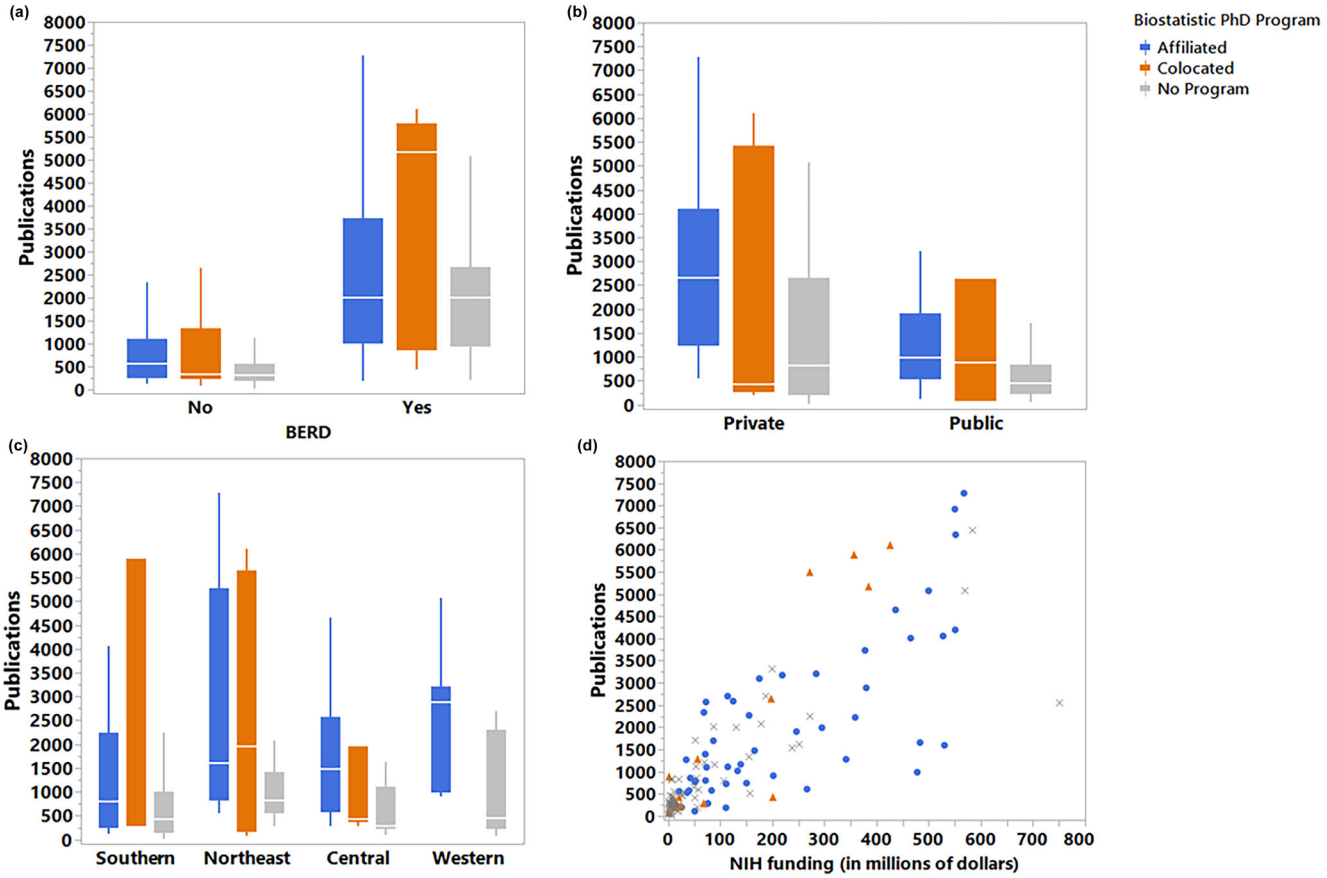


FIGURE 3 Box plot of publications for medical schools, stratified by affiliated (blue), collocated (orange), or no (grey) biostatistics PhD programme, based on presence of BERD programme (a), private versus public (b), and region (c) and scatterplot of publications for medical schools versus NIH funding (d). Abbreviations: BERD, Biostatistics, Epidemiology, and Research Design; NIH, National Institutes of Health.

The benefits of BERD programmes for academic health centres are well documented (Hanlon et al., 2022; Rahbar et al., 2017; Rubio et al., 2011; Welty et al., 2013). A primary function of BERD programmes is to provide support for investigator-initiated projects to produce manuscripts and attain NIH funding. However, a BERD programme can provide support and expertise across the entire research lifecycle, being able

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to guide research question creation, protocol development and review, study design and results dissemination. Moreover, a BERD programme can provide key education and mentoring for clinical faculty to help their professional development as researchers. The specialized expertise that a BERD programme can provide is a key component of success at academic health centres (Welty et al., 2013). Furthermore, specialization can refer to types of studies (e.g., comparative effectiveness research and clinical trials) and methodologies (e.g., predictive modelling and imaging data) with which a given biostatistician is more experienced. Moreover, specialization can also refer to medical specialties where a biostatistician maintains a long-term relationship with clinical faculty for specific areas, thus becoming familiar with specialty-specific variables and processes. This can support formation of investigator-focused professional relationships, as opposed to project-based relationships (Rubio et al., 2011; Tsevat et al., 2014; Welty et al., 2013).

Despite the documented success of BERD programmes in providing comprehensive and effective statistical expertise, additional strategies for optimizing biostatistical support are needed for historically under-resourced schools (particularly schools that currently lack a BERD programme or a significant portfolio of NIH funding) and schools located in the biostatistics deserts described in our study. In a recent survey of collaborative biostatistics units, 21% included no NIH-funded cores, 14% received no grant funding from the NIH and the median team included only three total members (Hanlon et al., 2022). Small, agile teams of biostatisticians and other research faculty or staff can maximize their impact on medical schools' research productivity by (a) developing a model of shared responsibilities for research mentorship and research operations, (b) proactively engaging clinical faculty and trainees in research to develop the local pool of investigators and (c) leveraging team members' diverse academic backgrounds, which may include exposure to statistical methods through training in academic disciplines not traditionally associated with the biostatistician role (Tumin, Baumgarten, et al., 2022; Tumin et al., 2020).

In addition to biostatistical support, several other factors were associated with medical school publications in our study, including NIH funding and region. A study dating back four decades supported a strong association between publications and NIH funding (McAllister & Narin, 1983). More recent studies have confirmed that relationship and found that publications supported by NIH funding have a greater impact in terms of journal prestige and number of citations (Boyack & Jordan, 2011; Lewison & Dawson, 1998). Nonetheless, our unadjusted analysis indicated that a $50 million increase in annual NIH funding was correlated with a 30% increase in annual publications, equivalent to 242 publications for the median medical school. Further work may consider how different sources of intramural or extramural funding correspond to the quantity and quality of resulting publications and how these data can be used to inform budgeting for biostatistical and other research support based on targets for scholarly productivity. Interestingly, the aforementioned 1983 study also found an association between publications and medical school region, with fewer publications from Southern US-based schools (McAllister & Narin, 1983). Our study found similar regional differences in publication output, with the highest output from Northeastern US-based schools and lowest outputs from Southern and Western US-based schools, supporting the persistence of regional disparity in terms of academically productive medical schools.

Our conclusions are limited by some aspects of the available data and the analytic approach. The ASA registry of graduate programmes may not have included all eligible training programmes in statistics and biostatistics, resulting in overestimation of the proportion of medical schools located in a biostatistics desert. We also did not examine affiliation or colocation of graduate programmes in other disciplines that include significant training in biostatistical methods. Our ascertainment of publication outcomes shares the limitation of other studies using the Scopus database (Gray et al., 2023), including delayed indexing of some publications and the potential for misattributing author affiliations or undercounting publications in non-indexed outlets. While we focused on scholarly publication as a highly visible output of biostatistical support, we were unable to quantify other contributions of biostatistical faculty and staff at each medical school, such as contributions to teaching, funding and career development of other faculty. In our multivariable analysis, it is also possible that the inclusion of BERD programme and NIH funding independent variables may have led to overcontrolling the model, since graduates of local biostatistics programmes may have been the biostatisticians staffing BERD programmes and contributing to the success of each medical school's NIH funding applications.

Despite these limitations, our study provides a novel analysis of access to biostatistics resources across US medical schools and reveals that a large share of MD-granting medical schools is operating in biostatistics deserts, where the capacity to develop and grow biostatistics units may be limited by the lack of local graduate programmes. Our findings further suggest that structured biostatistics services, such as the BERD programmes, may represent the best opportunity for medical schools to leverage the local biostatistics workforce to support scholarly publication; however, they also point to the need for more research on how biostatistics units should best be organized at smaller, geographically isolated or historically under-resourced medical schools.

AUTHOR CONTRIBUTIONS

Terrie Vasilopoulos: Conceptualization; data analysis; writing—original draft; writing—review and editing. **Dmitry Tumin:** Conceptualization; data acquisition and management; writing—original draft; writing—review and editing.

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CONFLICT OF INTEREST STATEMENT

TV discloses her unpaid position as chair-elect of the Statistical Consulting section of the American Statistical Association. DT discloses salary support from the Kate B. Reynolds Charitable Foundation and Lilly Grant Office for unrelated research and quality improvement projects.

DATA AVAILABILITY STATEMENT

Source data and replication code are available from the authors upon request.

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SUPPORTING INFORMATION

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